# STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

(SECMS)

Salesforce-Based Academic Enrollment Management Application

## Project Report

This report documents the ideation, requirement analysis, project planning, Salesforce configuration, automation, approval, security, reporting, and dashboard implementation of SECMS.

**Technology:** Salesforce Lightning / Salesforce Flows / Custom Objects / Validation Rules / Approval Processes / Reports & Dashboards

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# 1. INTRODUCTION

## 1.1 Project Overview

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain transparency into course allocations and academic progress.

The solution enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management. With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that enrollment operations—from admission to course completion—are handled systematically and monitored through key indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

## 1.2 Purpose

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

• Accurate student data management

• Structured enrollment workflows

• Automated approvals and validations

• Transparent reporting and monitoring

# 2. IDEATION PHASE

## 2.1 Problem Statement

| PS | I am (Customer) | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | Enrollment Officer | manage student enrollments efficiently | enrollment data is scattered | systems are manual and unintegrated | frustrated and error-prone |
| PS-2 | Administrator | track enrollments and revenue | reporting is delayed | data is not centralized | uncertain in decision-making |

## 2.2 Empathy Map Canvas

**User Persona:** Enrollment Officer

• **Think & Feel:** Concerned about double bookings and revenue loss

• **Hear:** Complaints from students and management

• **See:** Multiple spreadsheets and emails

• **Say & Do:** Rechecks data and follows up manually

• **Pain:** Duplicate entries and approval bottlenecks

• **Gain:** Automation, dashboards, and faster processing

## 2.3 Brainstorming & Idea Prioritization

**Step 1 – Brainstormed Ideas:** Spreadsheet automation; standalone enrollment portal; ERP integration; Salesforce-based enrollment system.

**Step 2 – Grouping:** Manual solutions; semi-automated solutions; fully automated cloud solutions.

**Step 3 – Prioritization:** Salesforce-based SECMS was selected based on the project's feasibility and expected impact.

# 3. REQUIREMENT ANALYSIS PHASE

## 3.1 Customer Journey Map

Stages covered: Awareness & Registration → Course Selection → Enrollment Request → Fee Payment → Approval → Instructor Assignment → Reporting & Completion.

## 3.2 Solution Requirements

| FR No | Epic | Description |
|---|---|---|
| FR-1 | Student Management | Manage student records |
| FR-2 | Course Management | Create and manage courses |
| FR-3 | Enrollment Processing | Request, approve, reject enrollments |
| FR-4 | Instructor Assignment | Auto-assign instructors |
| FR-5 | Reporting | Dashboards and reports |

## Non-Functional Requirements

| ID | Requirement | Description / Examples |
|---|---|---|
| NFR-1 | Usability | Simple, intuitive interface; quick registration; one-click approval; role-based dashboards. |
| NFR-2 | Security | Confidentiality and access control; admins modify course details; instructors see assigned students; students cannot access |
| NFR-3 | Reliability | Consistent enrollment processing; no duplicate registrations; accurate status updates. |
| NFR-4 | Performance | Fast dashboard refresh, student search, and bulk operations. |
| NFR-5 | Scalability | Support more students, courses, departments, and users; allow future modules and portal/mobile integration. |
| NFR-6 | Availability | Continuous access during institutional operations with minimized downtime and reliable cloud access. |

## 3.3 Data Flow Description

User inputs student and enrollment data. The system validates and stores records. Approval processes route requests to authorized approvers. Reports and dashboards consume processed data. The overall system validates data, routes approvals, assigns instructors, and generates reports.

## 3.4 Technology Stack

| Layer | Technology |
|---|---|
| UI | Salesforce Lightning |
| Logic | Salesforce Flows |
| Database | Salesforce Objects |
| Automation | Validation Rules, Approval Process |
| Reporting | Reports & Dashboards |

# 4. PROJECT DESIGN PHASE

## 4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, operational constraints, and automation using Salesforce-native capabilities.

## 4.2 Proposed Solution

| Parameter | Description |
|---|---|
| Problem | Manual enrollment handling |
| Solution | Salesforce-based SECMS |
| Novelty | End-to-end automation |
| Social Impact | Improved student experience |
| Revenue Model | Institutional deployment |
| Scalability | Multi-department support |

## 4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.

# 5. PROJECT PLANNING & SCHEDULING

## 5.1 Project Planning

Agile methodology is used with Epics → Stories → Story Points, sprint-based execution, and velocity-based estimation.

## Product Backlog and Sprint Schedule

| Sprint | FR / Epic | USN | User Story / Task | Points | Priority | Member |
|---|---|---|---|---|---|---|
| 1 | Developer Setup | USN-1 | Create Salesforce developer account for SECMS. | 1 | High | Member1 |
| 1 | Data Modeling | USN-2 | Create Student, Course, Enrollment, Instructor custom objects. | 3 | High | Member2 |
| 2 | Data Modeling | USN-3 | Create dedicated tabs for academic objects. | 3 | High | Member3 |
| 2 | Data Modeling | USN-4 | Create relevant student, course, and enrollment fields. | 5 | High | Member4 |
| 2 | Data Modeling | USN-5 | Create centralized Lightning App. | 3 | High | Member4 |
| 2 | Data Modeling | USN-6 | Define relationships between students, courses, and enrollments. | 5 | Medium | Member1 |
| 2 | Data Modeling | USN-7 | Customize page layouts. | 5 | High | Member2 |
| 3 | Automation | USN-8 | Validate mandatory fields and enrollment limits. | 3 | High | Member4 |
| 3 | Automation | USN-9 | Automate enrollment processing and notifications. | 3 | High | Member4 |
| 3 | Automation | USN-10 | Create approval process for enrollment requests. | 5 | High | Member1 |
| 4 | Security | USN-11 | Configure role-based security. | 5 | High | Member2 |
| 5 | Reports | USN-12 | Create reports on enrollments, registrations, assignments. | 4 | High | Member3 |
| 6 | Dashboards | USN-13 | Create visual enrollment statistics and trends. | 4 | Medium | Member4 |

## Project Tracker, Velocity & Burndown Chart

| Sprint | Points | Duration | Start | Planned End | Completed | Actual Release |
|---|---|---|---|---|---|---|
| Sprint-1 | 20 | 6 Days | 24 Oct 2022 | 29 Oct 2022 | 20 | 29 Oct 2022 |

| Sprint | Points | Duration | Start | Planned End | Completed | Actual Release |
|---|---|---|---|---|---|---|
| Sprint-2 | 20 | 6 Days | 31 Oct 2022 | 05 Nov 2022 | | |
| Sprint-3 | 20 | 6 Days | 07 Nov 2022 | 12 Nov 2022 | | |
| Sprint-4 | 20 | 6 Days | 14 Nov 2022 | 19 Nov 2022 | | |

# 6. PROJECT DEVELOPMENT PHASE

## Project Flow

• Milestone 1: Salesforce Developer Account Creation

• Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)

• Milestone 3: Tabs Creation

• Milestone 4: Lightning App Creation

• Milestone 5: Fields Creation

• Milestone 6: Create Relationships

• Milestone 7: Page Layout Customization

• Milestone 8: Validation Rules

• Milestone 9: Flows (Automation)

• Milestone 10: Approval Process

• Milestone 11: Reports

• Milestone 12: Dashboards

• Milestone 13: Security Setup

• Milestone 14: Conclusion

## What You Will Learn

• Real-time Salesforce project implementation

• Designing custom objects and relationships

• Creating page layouts and record types

• Writing validation rules

• Building automation with flows

• Creating approval workflows

• Developing reports and dashboards

• Setting up security with profiles, permission sets, and sharing

• End-to-end application testing

# 7. MILESTONE 1 – SALESFORCE DEVELOPER ACCOUNT

## Introduction

Salesforce is a cloud-based customer success platform used to manage relationships, processes, data, automation, analytics, and collaboration securely. For SECMS, Salesforce provides the platform for storing student, course, instructor, and enrollment data and automating the related workflows.

## Activity 1: Creating Developer Account

Create a developer org using the Salesforce Developer signup page. Enter first name, last name, email, role as Developer, college name as Company, India as Country, postal code, and a suitable username in the required format. Submit the signup form.

## Activity 2: Account Activation

Open the signup email, select Verify Account, create a password, answer the security question, and complete the account activation. The user is then redirected to the Salesforce Setup page.

# 8. MILESTONE 2 – CUSTOM OBJECTS

## Activity: Creating Student, Course, Instructor and Enrollment Objects

The purpose of custom objects is to store and manage the academic information required by SECMS. The project uses Student, Course, Instructor, and Enrollment objects.

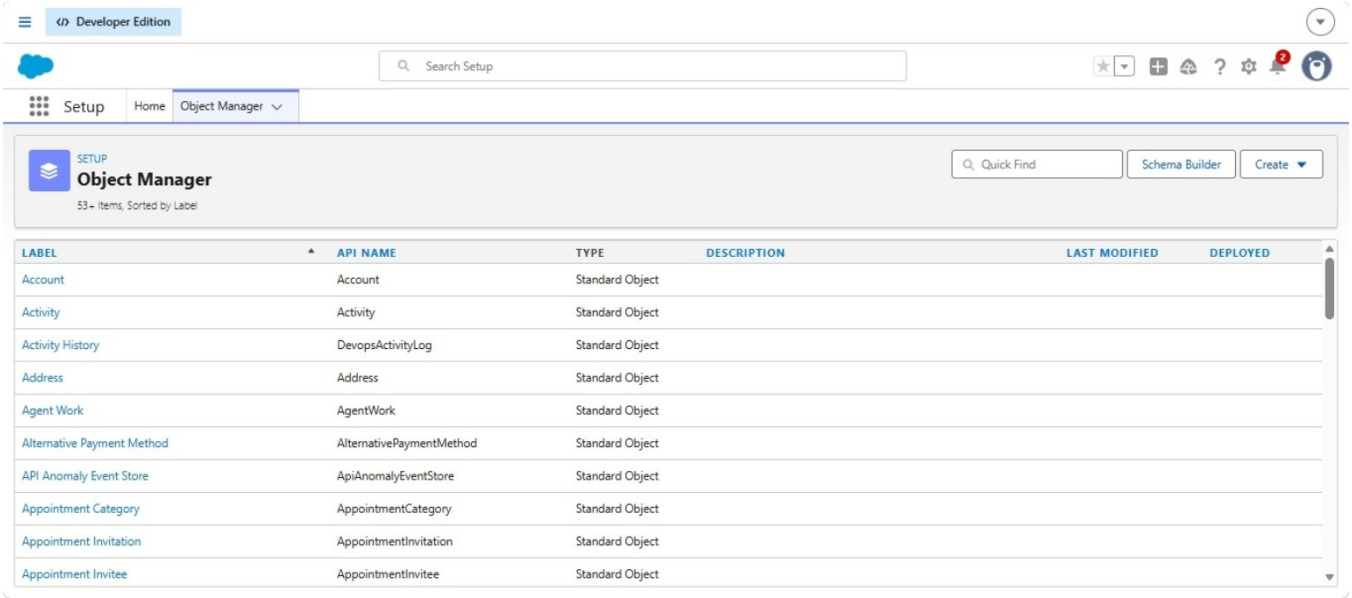



Figure 8.1 – Salesforce Object Manager

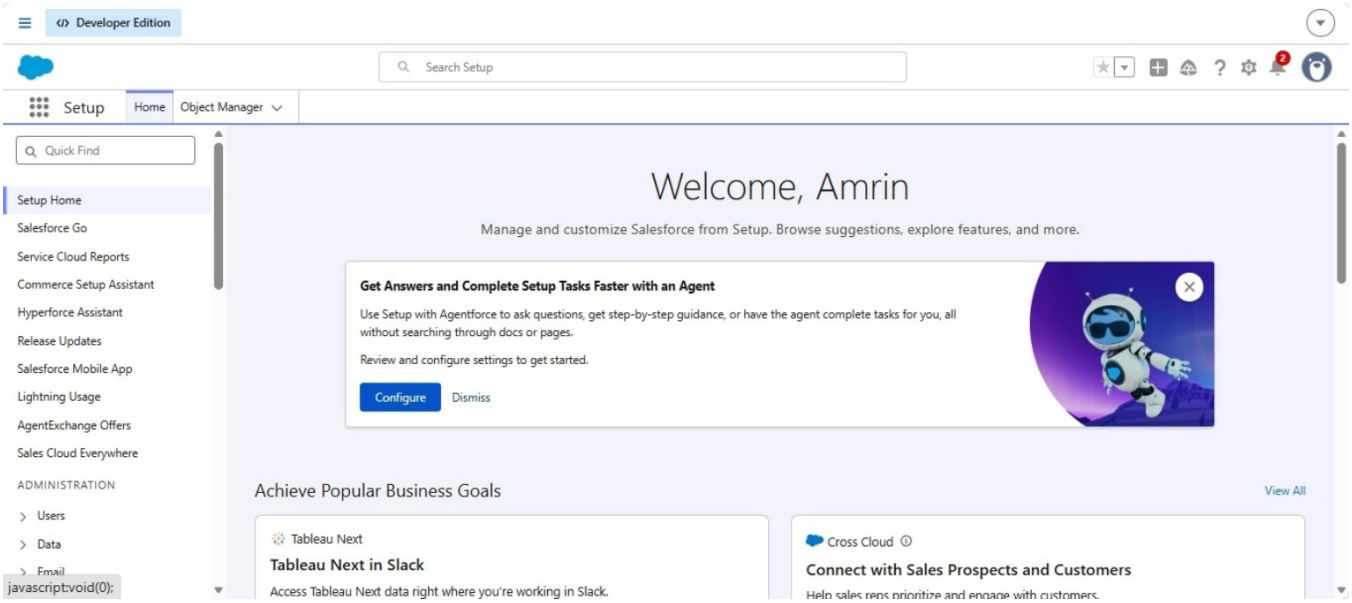



Figure 8.2 – Student object details

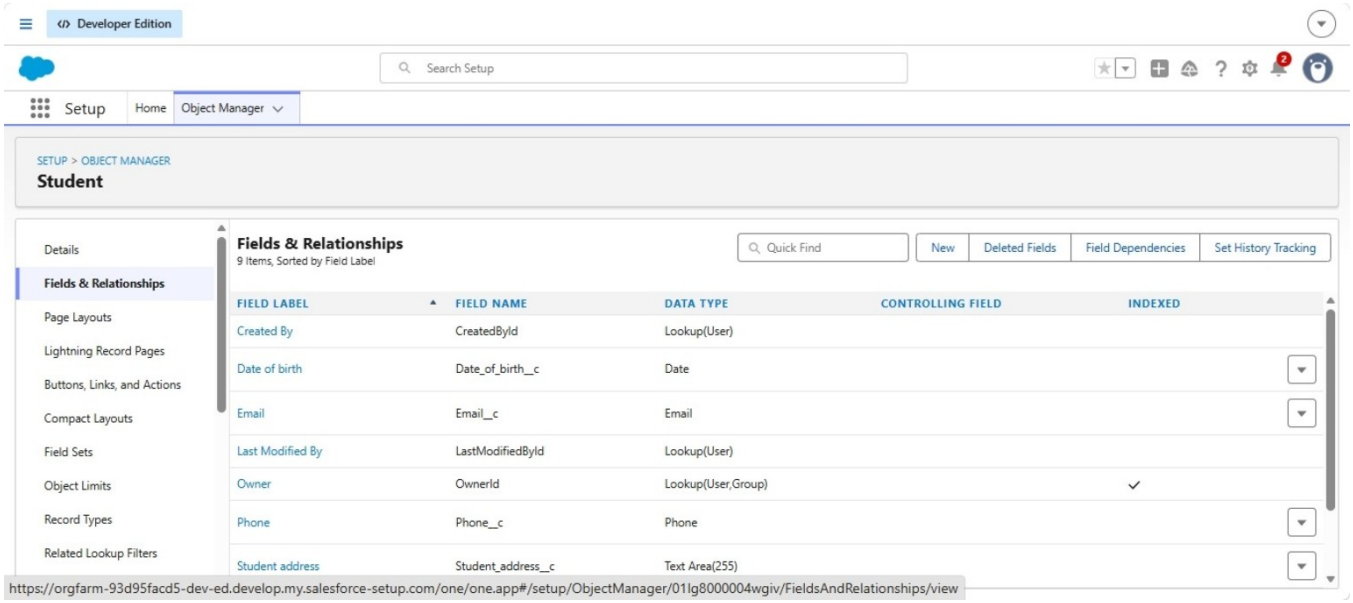



Figure 8.3 – Student fields and relationships

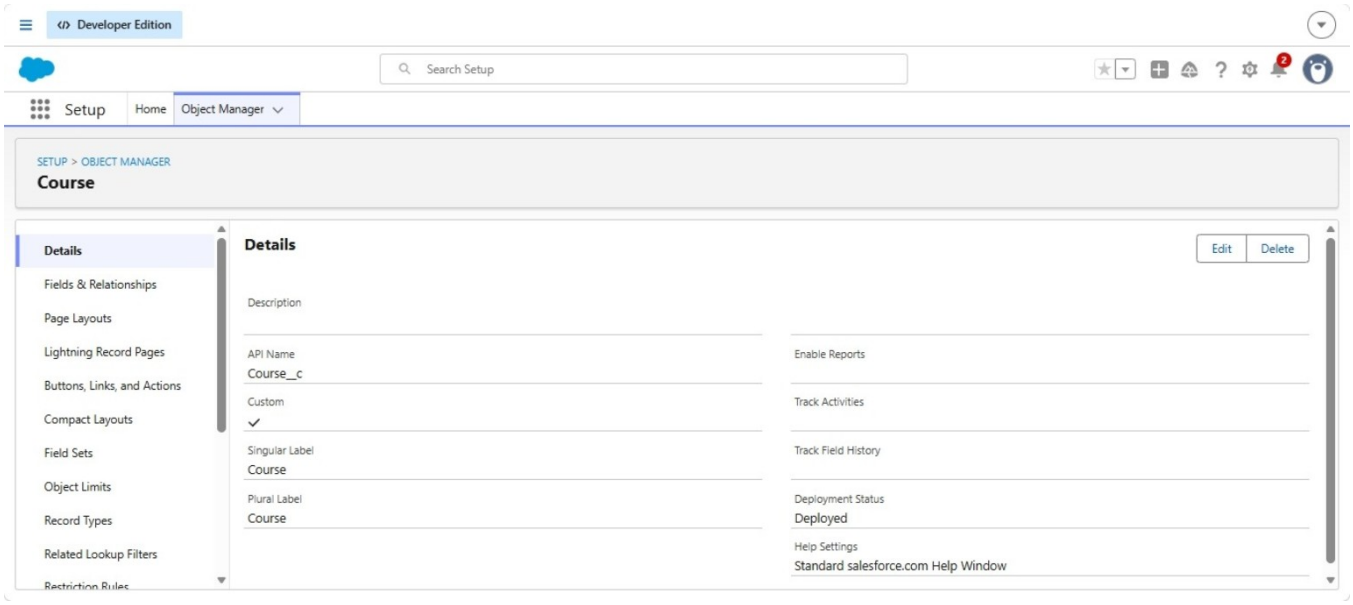



Figure 8.4 – Course object details

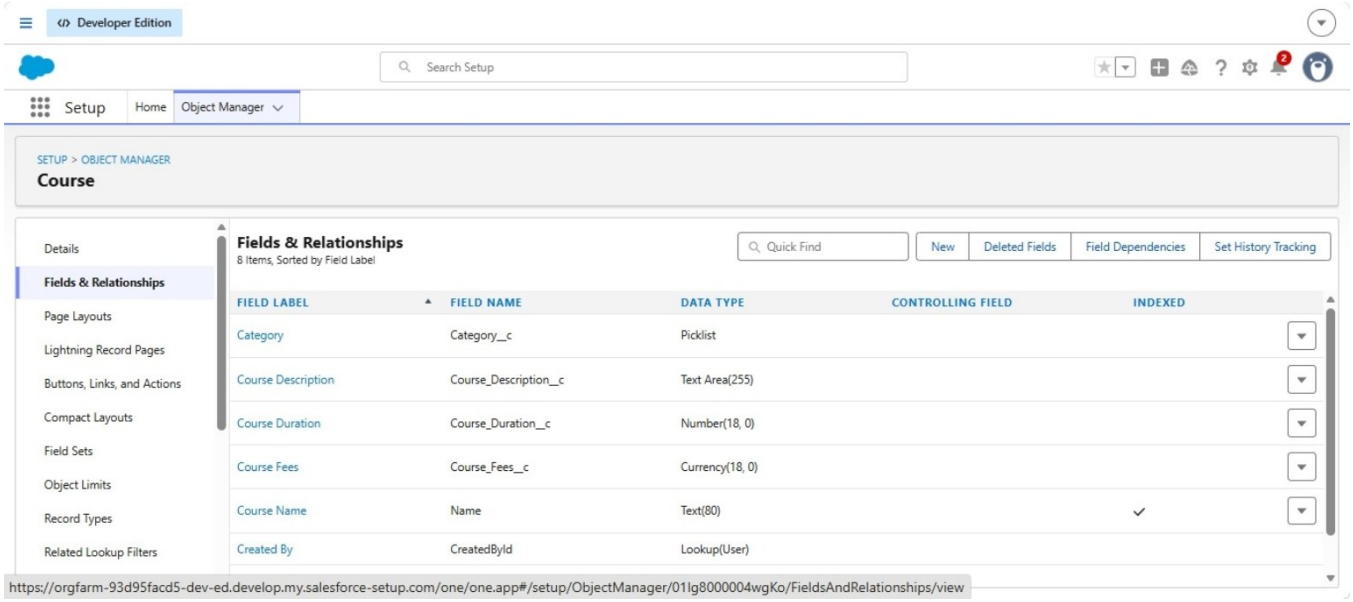



Figure 8.5 – Course fields and relationships

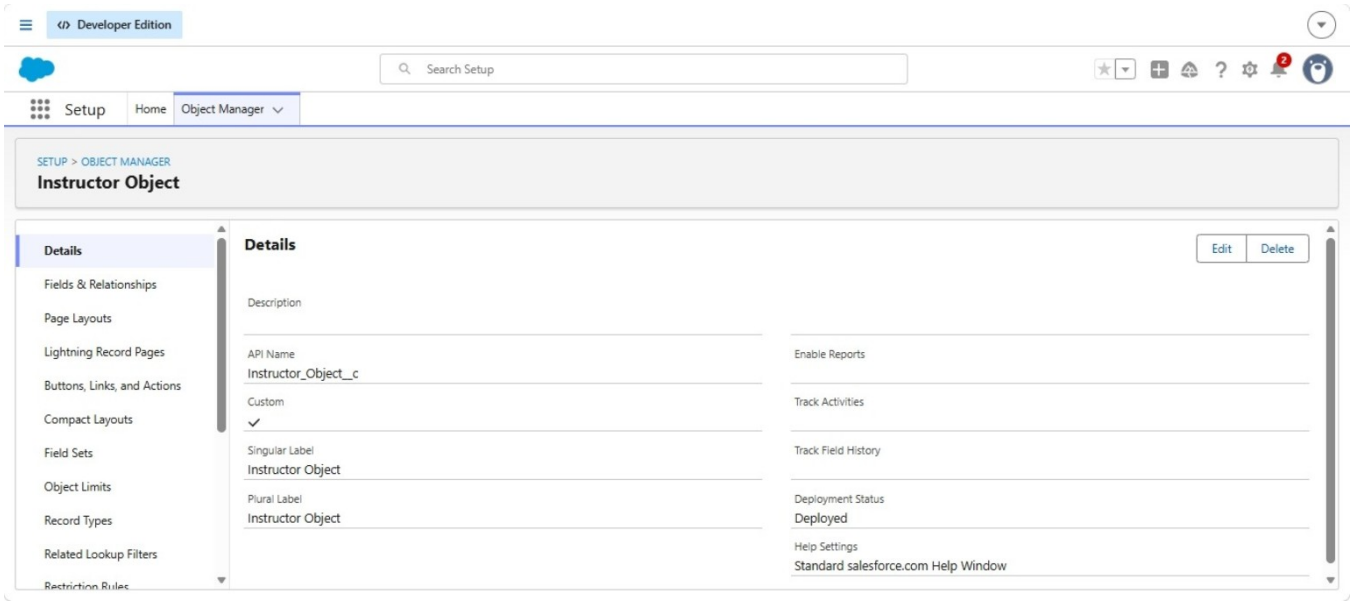



Figure 8.6 – Instructor object details

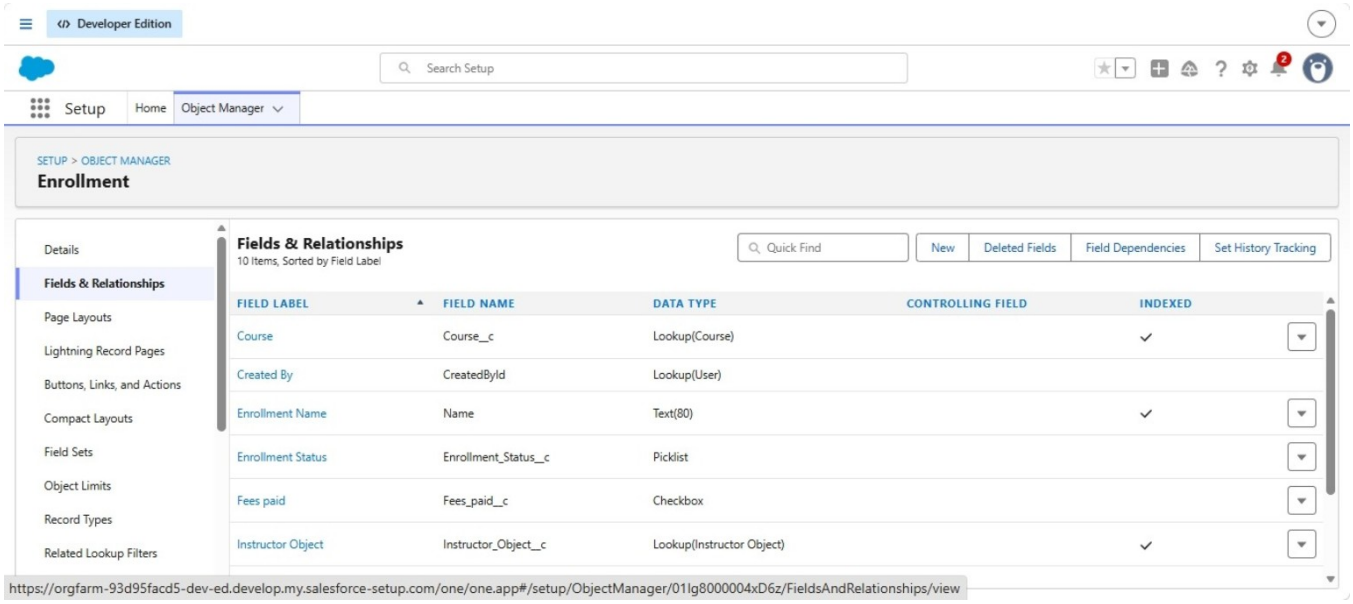



Figure 8.7 – Enrollment fields and relationships

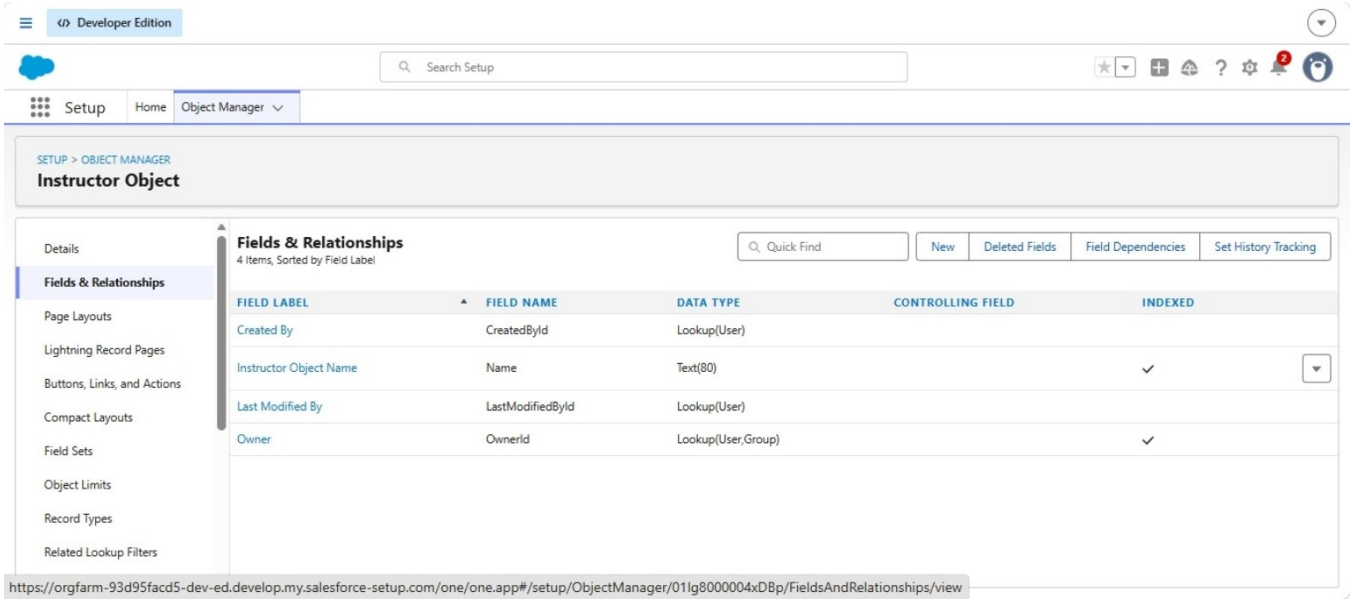



Figure 8.8 – Instructor object fields

# 9. MILESTONE 3 – FIELDS

## Table 1: Student Object Fields

| Field Label | Data Type |
|---|---|
| Student Name | Text (Standard) |
| Email | Email |
| Phone | Phone |
| Date of Birth | Date |
| Student Status | Picklist |
| Address | Text Area |

## Table 2: Course Object Fields

| Field Label | Data Type |
|---|---|
| Course Name | Text (Standard) |
| Category | Picklist |
| Duration | Number/Text |
| Course Fee | Currency |
| Description | Text Area |

## Table 3: Instructor Object Fields

| Field Label | Data Type |
|---|---|
| Instructor Name | Text (Standard) |
| Instructor Code | Text |
| Expertise | Picklist |
| Phone | Phone |
| Email | Email |

## Table 4: Enrollment Object Fields

| Field Label | Data Type |
|---|---|
| Student | Lookup (Student) |
| Course | Lookup (Course) |
| Instructor | Lookup (Instructor) |
| Enrollment Date | Date |
| Enrollment Status | Picklist |
| Fees Paid | Checkbox |
| Total Amount | Currency |
| Comments | Text Area |

# 10. MILESTONE 4 – CREATE RELATIONSHIPS

## Activity 1: Lookup from Enrollment to Student

Go to Setup → Object Manager → Enrollment → Fields & Relationships → New → Lookup Relationship. Select Student as the related object, use the field label Student, configure field-level security and page layout visibility, and save.

## Activity 2: Lookup from Enrollment to Course

Create a Lookup Relationship from Enrollment to Course using the Course field. This links each enrollment record to its selected course.

## Activity 3: Lookup from Enrollment to Instructor

Create a Lookup Relationship from Enrollment to Instructor using the Instructor field. This stores the instructor assigned to an enrollment.

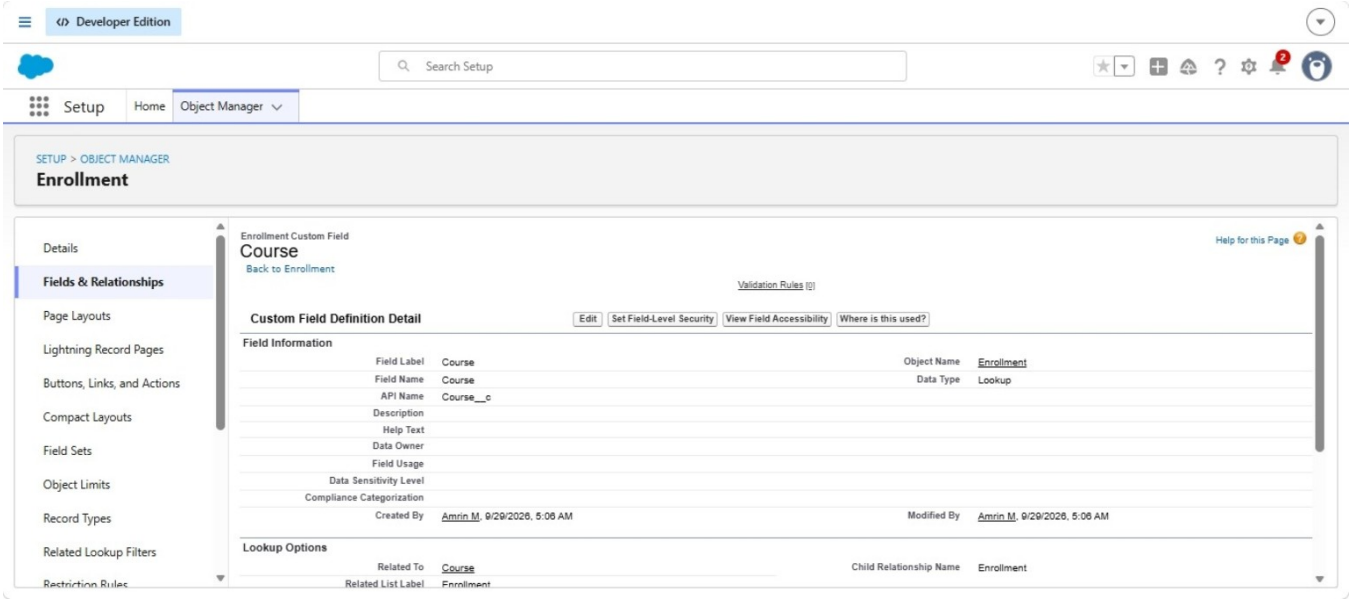



Figure 10.1 – Enrollment Course lookup field

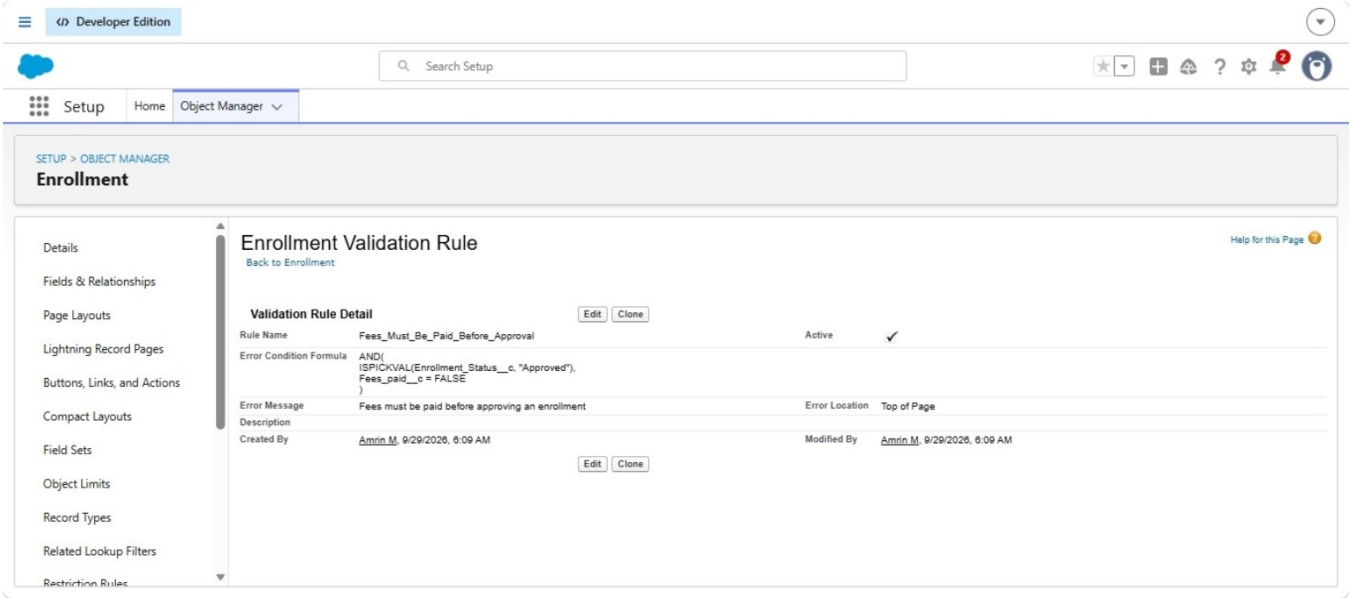



Figure 10.2 – Enrollment Instructor lookup field

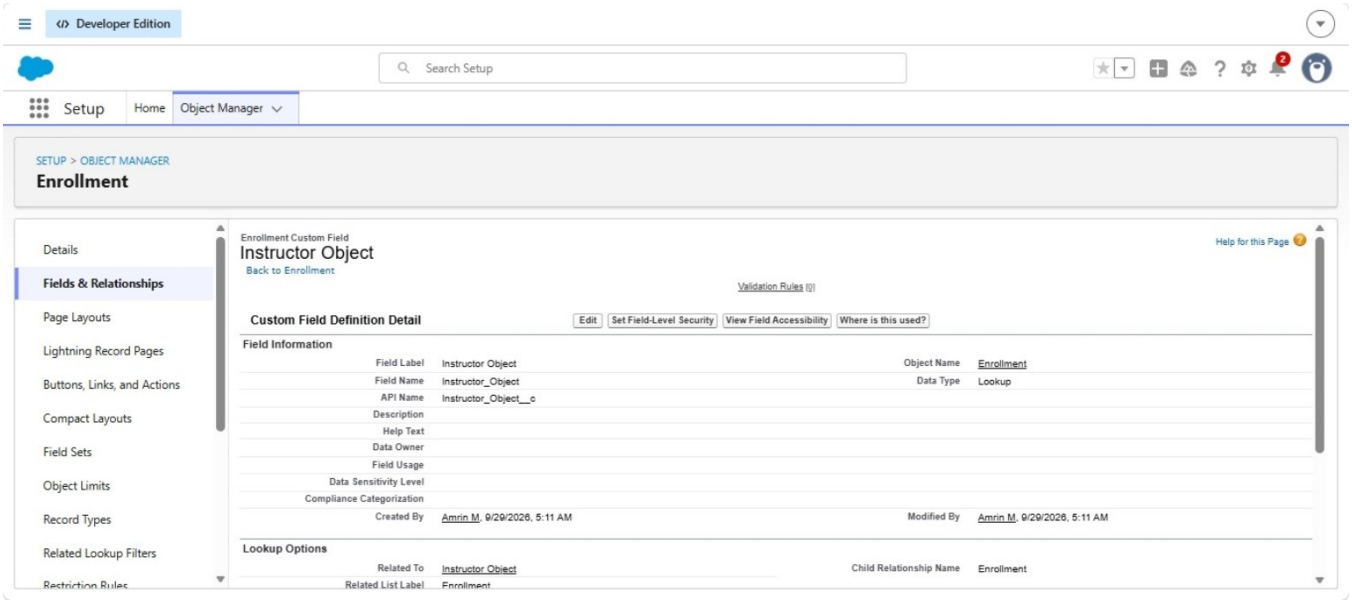



Figure 10.3 – Enrollment Student lookup field

# 11. MILESTONE 5 – VALIDATION RULES

## Activity 1: Student Email Domain Validation

Purpose: Ensure students use only allowed institutional email addresses. Example rule: NOT(CONTAINS(Email__c, "@student.edu")). Error message: Email must be a valid student institutional email.

## Activity 2: Fees Must Be Paid Before Approval

Purpose: Prevent an Enrollment from being approved until fees are paid. Formula: AND(ISPICKVAL(Enrollment_Status__c, "Approved"), Fees_paid__c = FALSE). Error message: Fees must be paid before approving an enrollment.

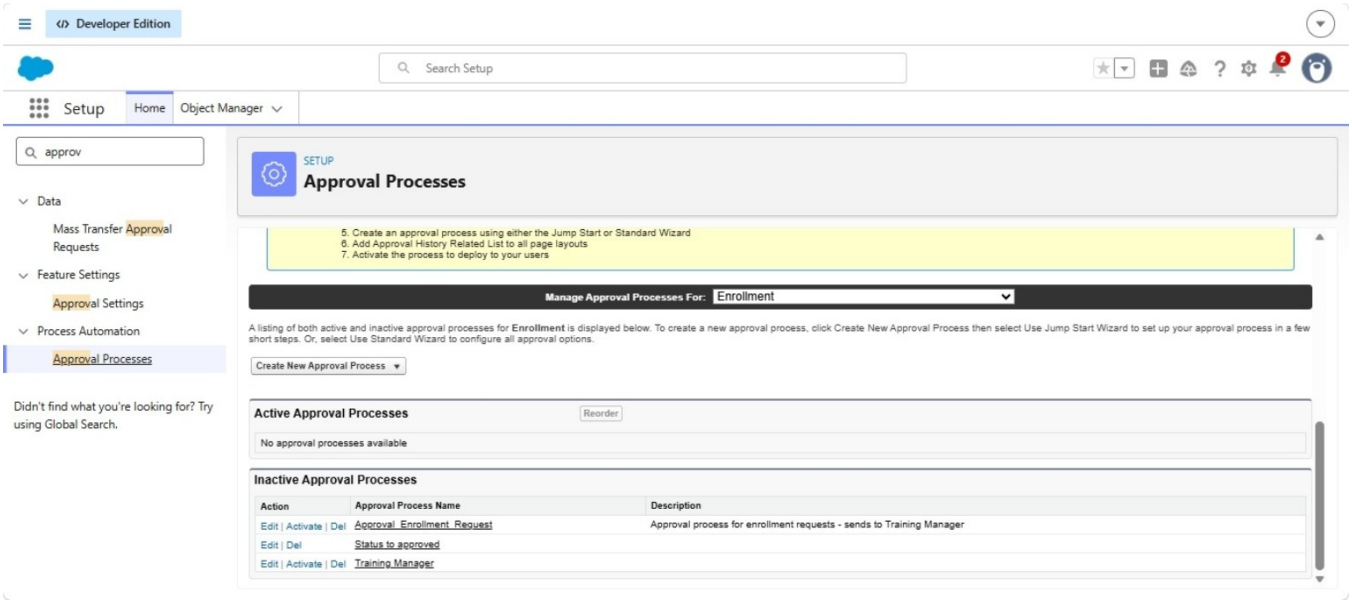



Figure 11.1 – Enrollment validation rule

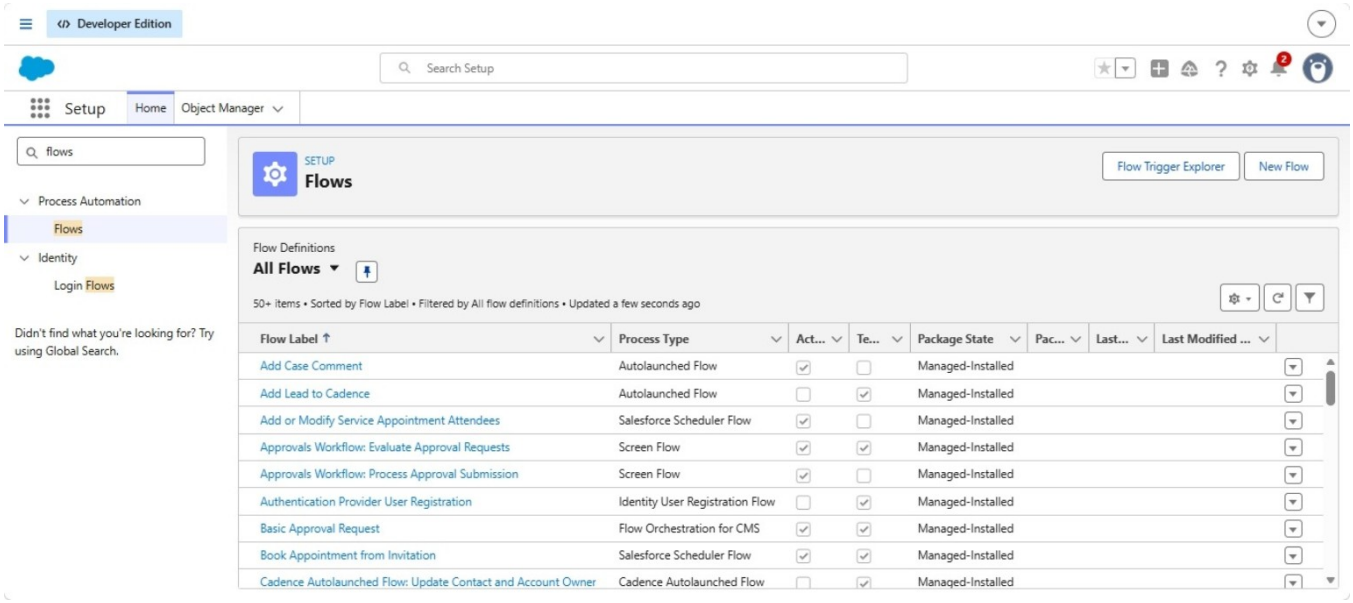



Figure 11.2 – Validation rule triggered when fees are not paid

# 12. MILESTONE 6 – APPROVAL PROCESS

## Activity: Create Approval Process for Enrollment Requests

Purpose: Route enrollment requests to the Training Manager for approval or rejection.

Configuration: Setup → Approval Processes → Enrollment. Create a new approval process using the standard setup. Process name: Approval_Enrollment_Request. Entry criteria: Enrollment_Status__c = "Requested". Add an approval step for the Training Manager. Final approval updates Enrollment_Status__c to Approved. Final rejection updates Enrollment_Status__c to Rejected and can send a rejection email. Activate the process.

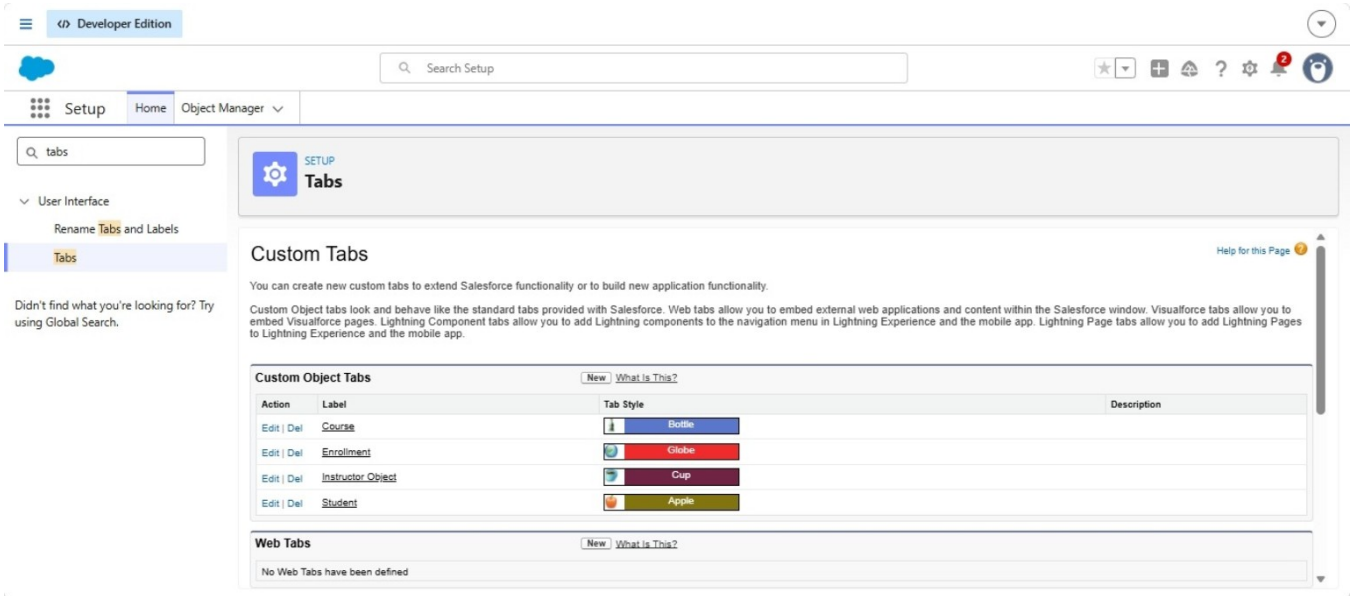



Figure 12.1 – Enrollment approval process configuration

# 13. MILESTONE 7 – TABS

Create custom tabs for Student, Course, Instructor and Enrollment from Setup → Tabs → Custom Object Tabs → New. Select the object, choose a tab style, assign profiles, and save.

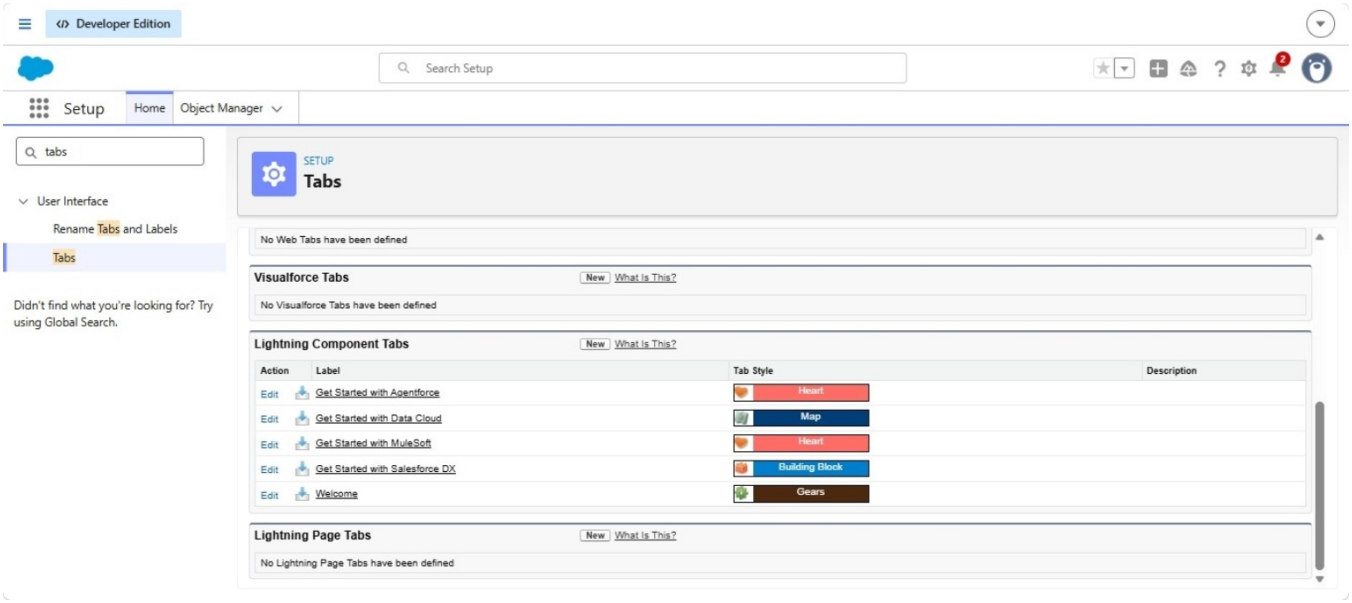



Figure 13.1 – Custom object tabs

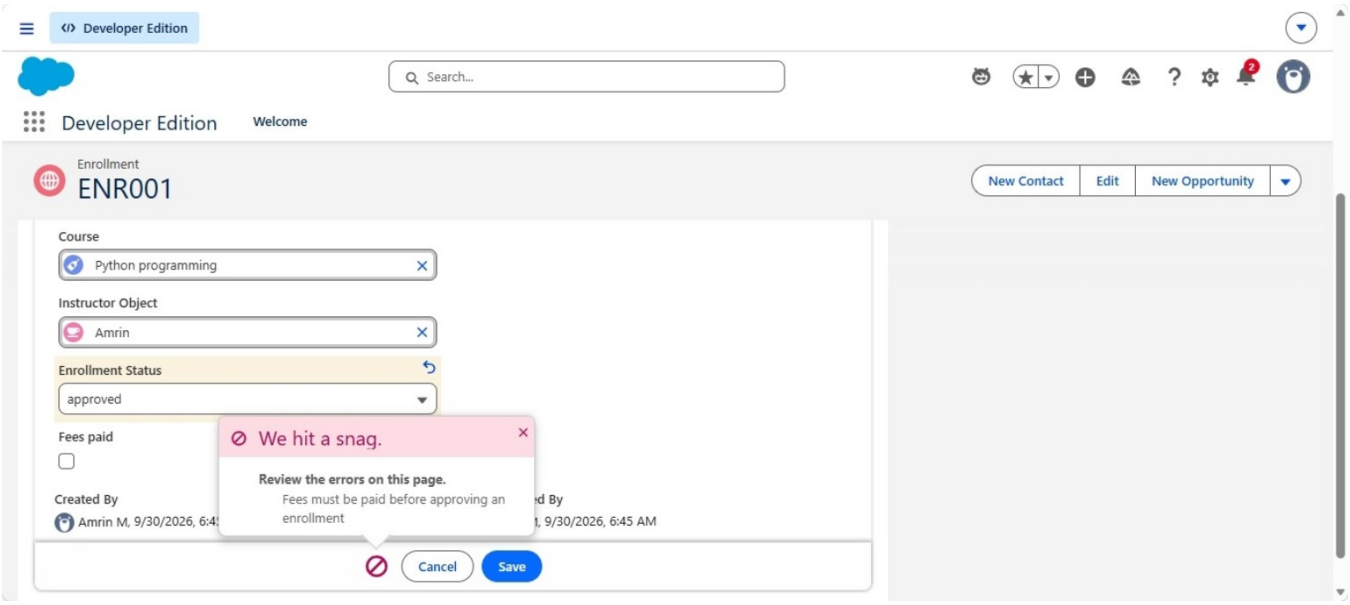



Figure 13.2 – Remaining tab sections

# 14. MILESTONE 8 – LIGHTNING APP & FLOWS

## Lightning App Creation

Navigate to Setup → App Manager → New Lightning App. App Name: Student Enrollment & Course Management. Use Standard Navigation. Assign the app to appropriate profiles and add Student, Course, Instructor, Enrollment, Reports, Dashboards, Home and Tasks to the navigation items. Review and save.

## Automation with Flows

Flows are used to automate enrollment processing, notifications, validations, and related actions. Create and activate the required flows from Setup → Process Automation → Flows.

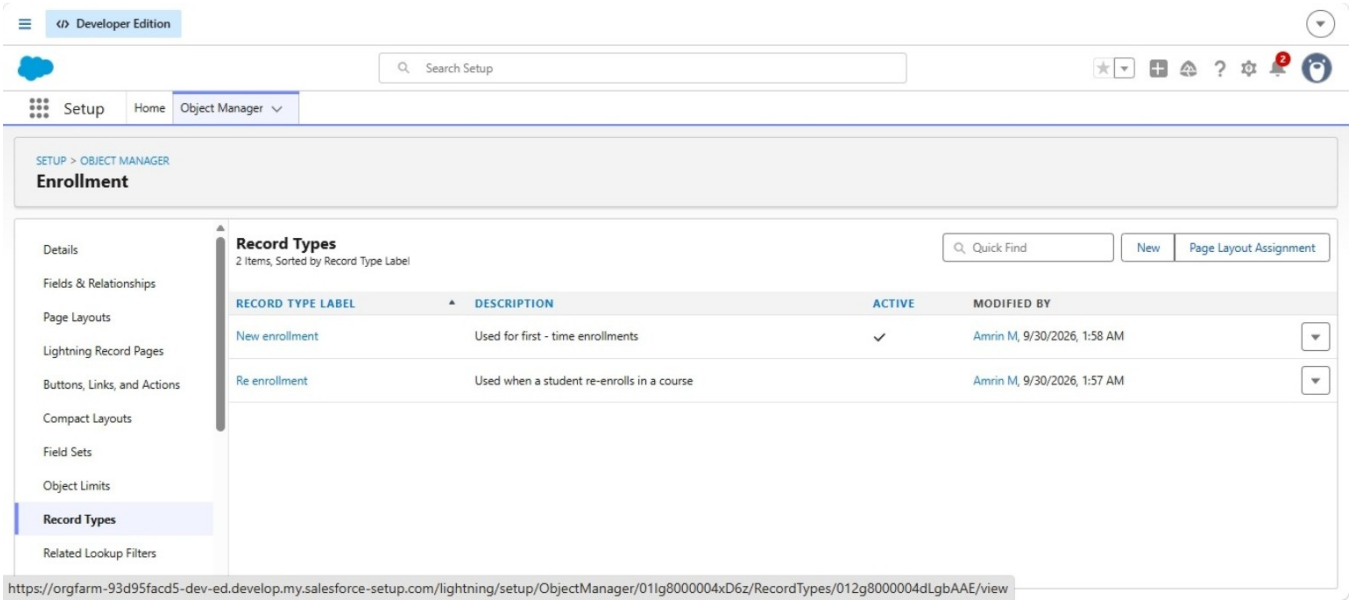



Figure 14.1 – Salesforce Flows page

# 15. MILESTONE 9 – PAGE LAYOUTS & RECORD TYPES

## Activity 1: Create Record Types for Enrollment

Create two Enrollment record types: New Enrollment and Re-Enrollment. New Enrollment is used for first-time enrollments. Re-Enrollment is used when a student enrolls again in a course. Enable the record types for the required profiles and assign appropriate default page layouts.

## Activity 2: Create Separate Page Layouts

Create New Enrollment Layout and Re-Enrollment Layout by cloning the existing Enrollment Layout. Keep the relevant fields for each process and remove unnecessary fields where appropriate.

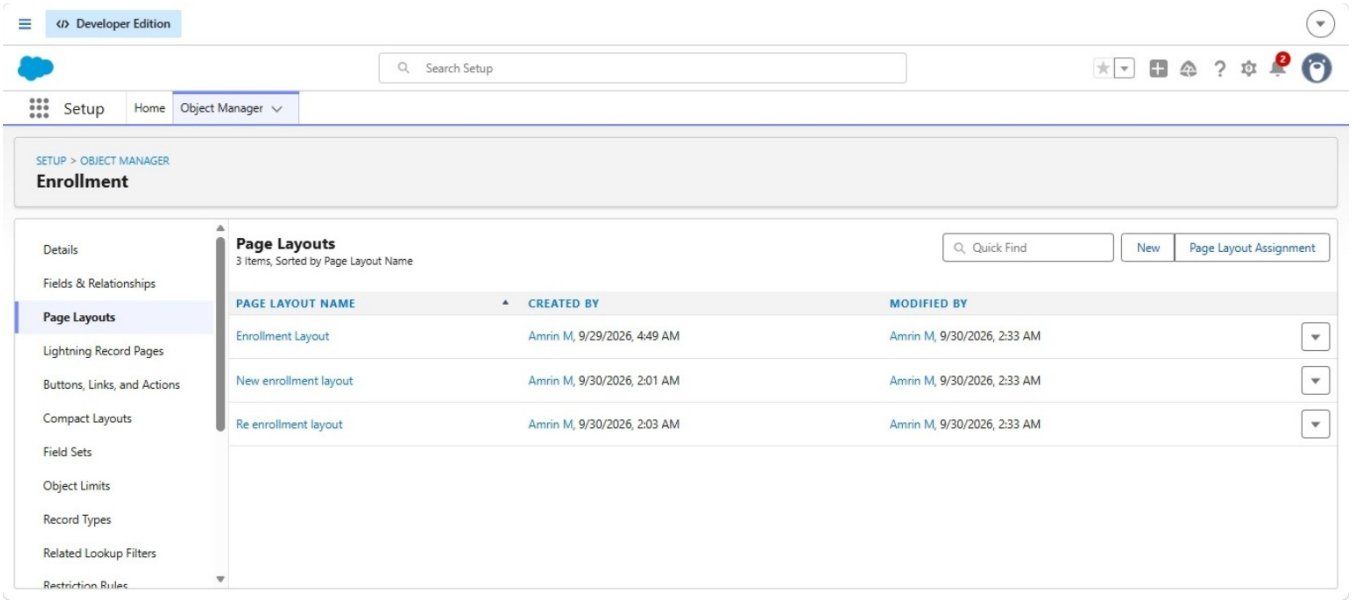



Figure 15.1 – Enrollment record types

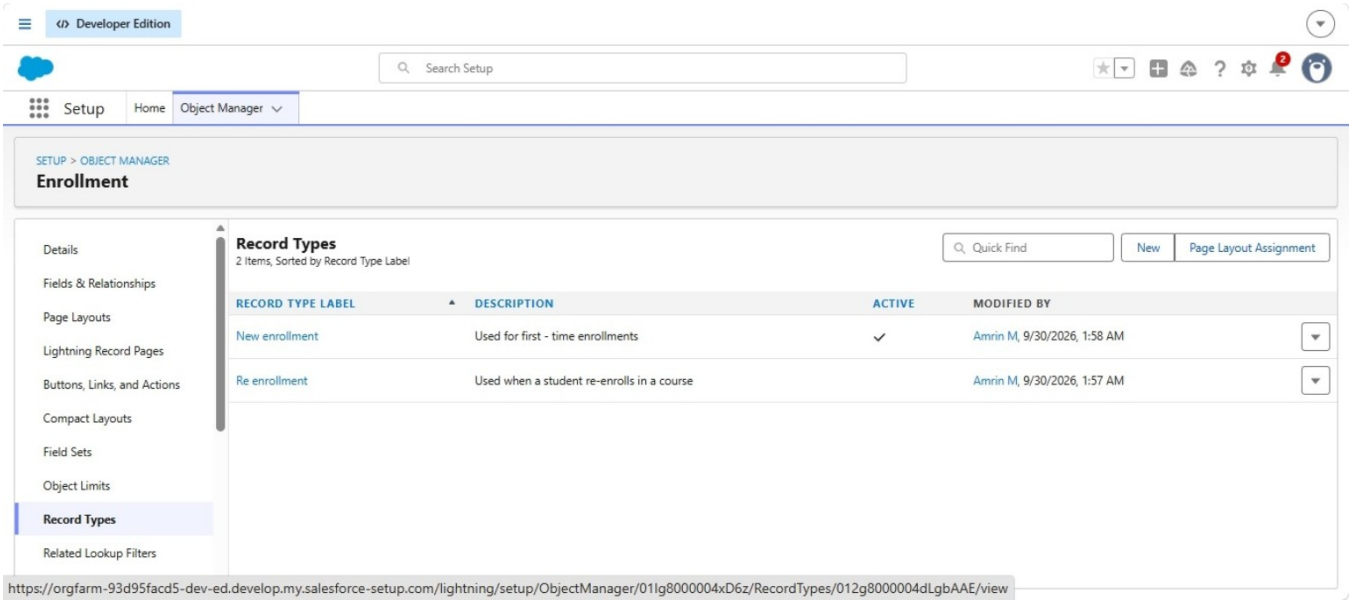



Figure 15.2 – Salesforce Flows used in the project

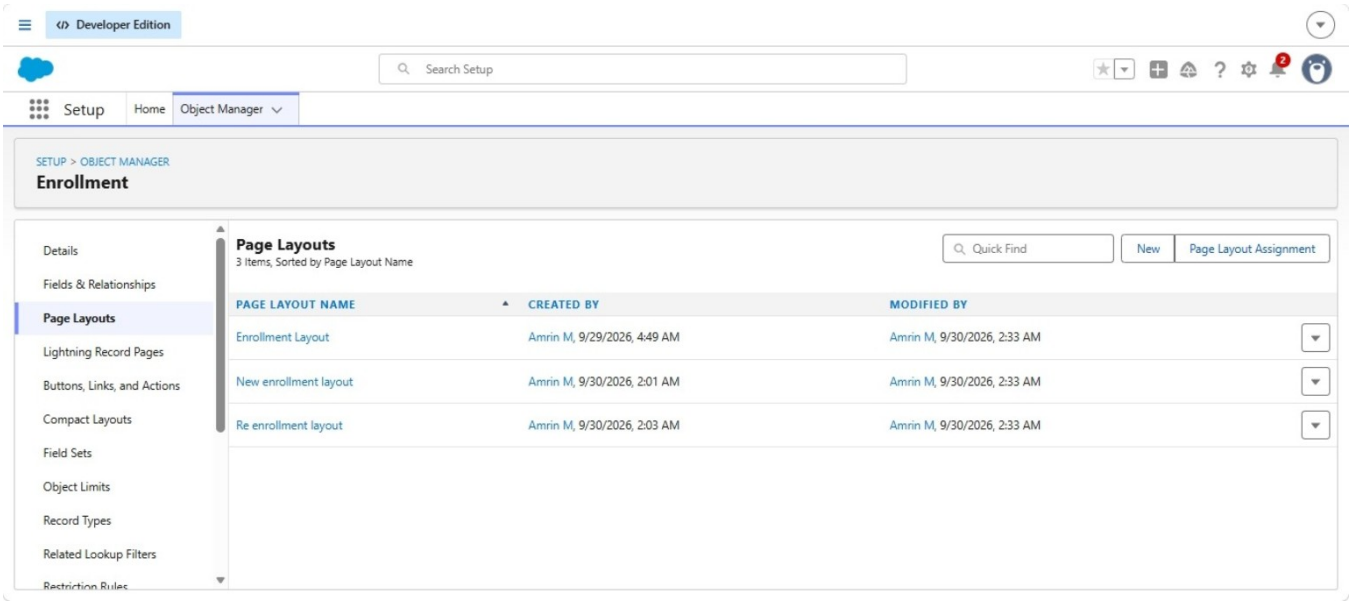



Figure 15.4 – Record type configuration (reference screenshot)

# 16. TESTING & VERIFICATION

The implemented configuration can be verified through functional tests covering object creation, field visibility, lookup relationships, validation rules, approval routing, record types, page layouts, tabs, flows, and user access.

| Test Case | Expected Result |
|---|---|
| Create Student record | Student record is stored with valid details. |
| Enter invalid institutional email | Validation rule prevents save and displays an error. |
| Create Enrollment with Student/Course/Instructor | Lookup relationships store the selected records. |
| Approve enrollment with Fees Paid = false | Save is blocked by validation rule. |
| Approve enrollment with Fees Paid = true | Enrollment can proceed to approval and status update. |
| Submit Requested enrollment | Approval process routes the request to the configured approver. |
| Use New Enrollment / Re-Enrollment | Correct record type and page layout are available. |
| Open custom tabs | Student, Course, Instructor and Enrollment tabs are accessible. |

| Test Case | Expected Result |
|---|---|
| View reports/dashboard | Configured academic metrics can be monitored. |

# 17. CONCLUSION & FUTURE ENHANCEMENTS

## Conclusion

The Student Enrollment & Course Management System provides a centralized Salesforce-based solution for managing student information, course offerings, enrollment requests, instructor assignments, approvals, and reporting. By using custom objects, lookup relationships, validation rules, flows, approval processes, page layouts, record types, reports, dashboards, and security controls, the system reduces manual work and improves data consistency and visibility.

## Future Enhancements

• Student self-service portal for registration and enrollment tracking

• Online payment gateway integration

• Automated instructor assignment based on expertise and availability

• Email/SMS notifications for enrollment and approval status

• Advanced dashboards for revenue, enrollment trends, and course performance

• Mobile-friendly access for students and staff

• Integration with institutional ERP or learning-management systems

**End of Project Report**